

WOLAN LOGISTICS

CUSTOM APP — FULL FEATURE SPECIFICATION

Built for Kampala, Uganda | Powered by 1% Standard Operations

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This document defines the complete feature requirements for the Wolan Logistics custom mobile and web application. It covers all three platforms — Admin Panel, Driver App, and Merchant App — and includes the multi-hub expansion architecture for scaling to 20+ branches across Kampala and Uganda.

1. THE APPS — OVERVIEW

The Wolan system requires three connected applications:

App	Who Uses It	Platform
Wolan Admin Panel	Hub Manager, Ops Coordinator, Director	Web + Mobile browser
Wolan Driver App	Riders (Boda Boda)	Android + iOS
Wolan Merchant App	Partner shops at Pioneer Mall & beyond	Android + iOS

2. ADMIN PANEL — CORE FEATURES

2.1 Dashboard (Home)

- Live delivery counter — completed, failed, pending (this week / today / custom range)
- Online drivers count in real time
- On-time delivery rate (%)
- Average pickup-to-delivery time
- Average placement-to-delivery time
- Average order rating + driver rating
- Daily target progress bar (e.g. Goal: 150 | Done: 92)

2.2 Orders / Dispatch

- Create new order manually
- Auto-assign order to nearest available rider
- Manual rider assignment override
- Dispatch time window setting (e.g. 1 hour buffer)
- Order status flow:
 - Pending → Picked Up → At Hub → Out for Delivery → Delivered / Failed
- Batch orders — group multiple packages to same direction for one rider
- Zone-based sorting (CBD, Kawempe, Ntinda, Makindye, Nakawa, Rubaga)
- Package staging shelf display — Incoming / Inspection / Zone Sort / Ready for Rider / Out
- Failed delivery handling — customer not available, refused, wrong address
- Return-to-merchant flow with return fee applied

2.3 Unique Wolan Order ID & Package Tracking

This is the core upgrade. Every order generates two separate tracking identifiers:

- Rider Tracking ID — tracks the rider's phone GPS location
- Package Tracking ID — unique serial number linked to a physical GPS tracker tag on the package
 - If rider abandons the package, the package tracker still shows exact location

- Package tracker status is independent from rider app status
- Customer tracking link shows BOTH rider location AND package location
- Hub can see if rider and package are in different locations — red flag alert
- QR code on each package links to live tracking page
- Tracking link shareable via SMS to customer automatically on dispatch

2.4 Drivers / Rider Management

- Add new driver profile — name, phone, plate, NIN, stage, next of kin
- Upload rider documents — ID, logbook, chairman stamp
- Security bond (250,000 UGX) tracking per rider
- Rider check-in status: Available / On Delivery / Break / Offline
- Daily rider performance board — deliveries, on-time rate, complaints
- Fine & penalty log — deducted from bond or commission
- GPS tracker (Cat-M1) status per rider — active / inactive / tampered alert
- Driver response time setting (e.g. 7 minutes to accept before reassignment)
- Proof of delivery — photo required before order marked complete
- Auto-accept option (can be toggled per rider or globally)
- Daily payment summary per rider

2.5 Live Map

- 45-inch-optimised command display view
- All active riders shown on map with delivery status
- Package tracker pins shown on map separately from rider pins
- Zone overlay — colour-coded delivery zones
- Delayed rider highlighted in red
- Idle riders flagged if stationary for more than 10 minutes
- Click rider to see current order, destination, ETA

2.6 Merchant Management

- Add merchant profiles — name, shop, building, phone, email
- Merchant QR code generated and printable for each shop
- Merchant login portal — they can place orders, track deliveries
- Real-time order visibility per merchant
- Merchant referral code system — M2M (Merchant to Merchant)
- Referring shop earns 50 UGX per delivery from referred shop
- Merchant level system: Starter / Active / Priority / Elite (based on volume)
- COD handling per merchant — fee calculation, payout tracking

2.7 Notifications

- New order alert to ops coordinator
- Rider gone dark alert (GPS inactive 15+ minutes)

- Tampered seal alert
- Broken package chain-of-custody alert
- Package and rider location mismatch alert
- COD limit alert — when rider approaches 1,000,000 UGX threshold
- Daily summary report push notification to Hub Manager

2.8 Customer Notifications

- Automated SMS/WhatsApp on dispatch — includes tracking link
- Tracking link shows: rider name, phone, live map, ETA
- OTP delivery confirmation — 4-digit code, rider enters in app
- Delivery receipt sent to customer and merchant on completion
- Delay alerts if order runs late
- Feedback/rating request sent 24 hours after delivery

2.9 Reports

- Sales report — daily, weekly, monthly revenue
- Driver performance report — deliveries, on-time rate, earnings
- COD reconciliation report — cash collected vs digital records
- Zone heatmap — where deliveries are concentrated
- Customer report — order counts, spending, retention
- Refund and failed delivery report
- Extended report — all order-level details exportable
- Analytics — time-based charts and trends

2.10 Settings

- Business settings — name, logo, service times, max delivery time
- Brand customization — colours, tracking handle, SMS template
- Dispatch settings — auto-assign toggle, dispatch time window
- Driver settings — response time, proof of delivery, item check
- Customer notification toggles — email, SMS, WhatsApp
- Location settings — city, timezone, distance unit (km)
- User/team management — invite ops staff with roles
- Currency — UGX

3. DRIVER APP — FEATURES

3.1 Login & Status

- Login with phone number + PIN
- Set status: Available / On Delivery / Break / Offline

- GPS tracker must be on — app warns if inactive

3.2 Order Flow

- Receive order notification — accept or reject within set time
- View order details — customer name, phone, address, package value
- View earnings before accepting
- Scan package QR code at hub to confirm custody transfer
- Navigate to delivery address — in-app map
- Call customer directly from app
- Mark as delivered using OTP entered by customer
- If customer unavailable — mark and return with photo proof
- Photo proof of delivery required before completion

3.3 Package Tracker Scan

- Rider must scan physical package GPS tracker tag before leaving hub
- This links the package tracker ID to the active delivery
- If package tracker and rider GPS diverge by more than 500 metres — alert fires

3.4 Earnings & Payments

- Daily earnings summary
- Bonus progress — e.g. 20-hit bonus, 35-sprint bonus
- Fine deductions visible
- COD amount carried — warning when approaching 1M UGX limit

3.5 Incident Reporting

- Report: accident / theft / mechanical failure / police stop
- Must report within 5 minutes — ops notified instantly
- Photo upload for incident

4. MERCHANT APP — FEATURES

4.1 Order Placement

- Tap QR code or log in to open order form
- Enter: customer name, phone, delivery address, item description, declared value
- Order submitted — unique Order ID generated instantly
- Secret code displayed for pickup agent identity verification
- Press 'Send Off' when pickup agent collects the item

4.2 Order Tracking

- Live status of each order — from pickup to delivery
- See rider location and package location (dual tracking)
- Delivery confirmation with timestamp, OTP proof, rider ID

4.3 COD & Payments

- View COD due amounts
- Payout status and history
- Commission and referral earnings visible

4.4 Merchant Dashboard

- Total deliveries this month
- Delivery level / tier status (Starter / Active / Priority / Elite)
- Referral code and referred shops count
- Earnings from M2M referral bonuses

5. PACKAGE GPS TRACKING — TECHNICAL SPEC

This is Wolan's most important differentiation from all other local logistics providers. Every package carries its own GPS identity, independent of the rider.

Feature	How It Works
Physical Tag	Small Cat-M1 GPS tracker sealed inside or attached to Wolan security bag
Unique Tag ID	Each tag has a serial number, linked to Order ID on scan at hub
Dual Tracking	Customer sees rider pin AND package pin on the same tracking page
Mismatch Alert	If package and rider are 500m+ apart, control room gets instant alert
Package Status	Tag shows: At Hub / In Transit / Stationary / Not Moving (possible tamper)
Lost Package	If rider goes dark, package tracker still shows last known GPS location
Tag Reuse	Tag is recovered at delivery, reset, and reused for next order

6. MULTI-HUB ARCHITECTURE — 20+ BRANCH EXPANSION ★ NEW

This section defines how the Wolan app must be architected to support expansion from 1 hub (Pioneer Mall) to 20+ hubs across Kampala and Uganda, without data collision or operational conflicts.

6.1 The Core Concept — Multi-Tenant Architecture

The system must use **Multi-Tenant Architecture**. This means one single app and one single server supports all hubs simultaneously, but each hub operates in complete isolation. Think of it like MTN Uganda — one network,

many towers across the city, one headquarters sees everything.

- Every record in the database (orders, riders, merchants, packages, COD) is tagged with a Hub ID
- Pioneer Mall Hub sees ONLY Pioneer Mall orders and riders
- Nakawa Hub sees ONLY Nakawa orders and riders
- No hub can view, edit, or interfere with another hub's data
- All hubs share the same codebase — no separate apps needed per hub

6.2 Hub ID Database Tagging

Every table in the database must include a **hub_id** foreign key. Every query filters by **hub_id** automatically based on the logged-in user's assigned hub.

Record Type	Hub Tag Example
Orders	order_id: #00234 hub_id: HUB_001 (Pioneer Mall)
Riders	rider_id: R_045 hub_id: HUB_003 (Nakawa)
Merchants	merchant_id: M_120 hub_id: HUB_002 (Ntinda)
COD Records	cod_id: C_089 hub_id: HUB_001 (Pioneer Mall)
Incidents	incident_id: I_012 hub_id: HUB_005 (Makindye)

6.3 The Three Dashboard Levels

Level	Who	What They See
Hub Dashboard	Hub Manager, Ops Coordinator	Their hub only — orders, riders, COD, map
Regional Dashboard	COO / Regional Manager	Assigned group of hubs (e.g. all CBD hubs)
HQ Master Dashboard	Director, General Manager	ALL hubs in real time — full overview

6.4 HQ Master Dashboard — Requirements

- Total deliveries across ALL hubs — live counter
- Per-hub delivery count — side-by-side comparison
- Best performing hub today — highlighted
- Lowest performing hub — flagged for attention
- Total COD in the field across all hubs combined
- Total revenue today / this week / this month — all hubs
- Map view — all rider pins from all hubs on one city-wide map
- Drill down — click any hub to enter its isolated dashboard
- Add a new hub — create Hub ID, assign manager, set zone — live in minutes
- Suspend a hub — freeze all operations at one hub without affecting others

6.5 Hub Manager Dashboard — Isolation Rules

- Sees their hub's live delivery map only
- Their hub's riders and merchants only
- Their hub's daily COD reconciliation
- Their hub's performance vs. daily target
- Cannot see or access any other hub's data — by design

6.6 Opening a New Hub — Process Flow

Opening Hub 2 must require zero new code. The process:

- Step 1 — Director logs into HQ Master Dashboard
- Step 2 — Click 'Add New Hub'
- Step 3 — Enter hub name, location, zone coverage
- Step 4 — System generates Hub ID automatically (e.g. HUB_002)
- Step 5 — Invite Hub Manager via phone/email — they get login instantly
- Step 6 — Hub is live and isolated from all other hubs

6.7 Inter-Hub Relay — Phase 3

In Phase 3, hubs will support cross-hub delivery relay:

- Transfer request sent from Hub A to Hub B for last-mile handoff
- Hub B accepts or rejects the relay
- Package custody transfers digitally — chain of custody maintained
- Both hubs see the package during transit
- Director sees full relay journey on HQ map

7. COD & FINANCIAL CONTROLS

- 10% COD service fee auto-calculated per order
- Max COD per rider: 1,000,000 UGX — system blocks new assignments beyond limit
- Max total COD in field: 5,000,000 UGX — dashboard alert
- Daily COD reconciliation — rider must settle before end of shift
- COD paid via Mobile Money to merchants — daily at 8:00 PM
- All transactions logged in admin panel
- Package insurance — 2,000 UGX fee, covers up to 500,000 UGX at 50% payout

8. CRISIS & SECURITY PROTOCOLS

- Rider gone dark — GPS off 15 minutes triggers field response alert
- Broken seal — rider flagged, investigation record opened
- Package and rider location mismatch — instant control room alert
- Manual mode — printed waybill book + SMS fallback when internet fails

- Hub security — only 3 roles can touch packages: Pickup Agent, Ops Coordinator, Rider
- Lost package — rider account frozen, scan history reviewed, merchant contacted
- Damage liability flow — merchant / Wolan / rider responsibility classification

9. FUTURE PHASES

Feature	Phase
Route planning & optimisation per rider	Phase 2
Geofencing for pickup and drop-off zones	Phase 2
WhatsApp notifications	Phase 2
Delay alerts auto-sent to customer	Phase 2
Multi-user team access with roles	Phase 2
Inter-hub relay dispatch	Phase 3
Wolan Locker System (National Grid)	Phase 3
Dark Store / warehousing module	Phase 3
20+ Hub full expansion	Phase 3

10. CONFIRMED TECH STACK

Layer	Technology	Purpose
Mobile Apps	React Native	Android + iOS — Driver and Merchant apps
Admin Panel	React (web)	Browser-based command center
Backend	Node.js or Python FastAPI	API, business logic, hub routing
Database	PostgreSQL	All records with hub_id on every table
Real-time	WebSockets	Live map, counters, alerts — no page refresh
Maps	Google Maps API	Rider + package tracking, zone display
Package GPS	Traccar + Mictrack MT700 / Teltonika FMM130	Cat-M1 physical package tracker
Payments	Flutterwave	MTN Mobile Money + Airtel Money
SMS	Africa's Talking	Uganda MTN / Airtel SMS
WhatsApp	WhatsApp Business API	Order status updates

Layer	Technology	Purpose
Email	AWS SES	Delivery receipts
Hosting	AWS Cape Town	Low latency for East Africa

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